# Response to Reviewers

*Estimating the causal effect of replying "👍" versus "merci" to professional emails*

We thank both reviewers, and the editor, for their careful reading of the manuscript. Below we address each comment in turn, with the corresponding change cited by page number in the manuscript.

## Reviewer 1

**Reviewer 1 — comment 1**
*"Sign-off" needs an operational definition — in particular, please clarify whether a bare emoji reaction counts as one.*

We have added an explicit definition, following the corpus-based taxonomy of Cc and Bcc, which does include bare emoji reactions.

**p. 1** — We define a sign-off , following the corpus-based taxonomy of [1], as any content appended after the substantive body of an email and before the sender's name, including a bare emoji reaction sent as the entirety of a reply.

**Reviewer 1 — comment 2**
*A simple difference in proportions across sign-off groups does not account for confounding — recipient seniority and email length both plausibly affect both sign-off choice and reply probability.*

Agreed. We now use inverse probability weighting, with weights from a propensity score model for sign-off choice conditional on recipient seniority, email length, and day of week sent.

**p. 2** — ~~The reply rate was compared across sign-off groups using a simple difference in proportions.~~ The reply rate was compared across sign-off groups using inverse probability weighting [2], with weights derived from a propensity score model for sign-off choice conditional on recipient seniority, email length, and day of week sent.

**Reviewer 1 — comment 3**
*Please also update the sign-off assignment model itself to match the revised analysis.*

Done; sign-off choice is now modeled explicitly as a logistic function of the covariates listed above. (modified on p. 2)

**Reviewer 1 — comment 4**
*Were message bodies actually read by the study team, or only metadata? This has ethical as well as methodological implications.*

Only metadata; we have kept this wording, which we believe is already explicit on this point.

**p. 2** — Metadata were extracted automatically from email headers and timestamps; message bodies were not read by the study team.

**Reviewer 1 — comment 5**
*Table 1's "recipient is senior" figure for the "merci" group looks inconsistent with the number reported in the abstract — please reconcile.*

Thank you for catching this; the table has been corrected.

**p. 2** —

| | "👍" | "merci" | Control |
|---|---|---|---|
| Recipient is senior, % | 31 | ~~42~~ 47 | 38 |
| Email length, mean words | 58 | 64 | 61 |
| Sent on Friday, % | 22 | 19 | 24 |
| Sender battery level, mean % | 34 | 71 | 68 |

Table I: Baseline characteristics by sign-off group.

**Reviewer 1 — comment 6**
*The calibration equation appears to duplicate the sign-off assignment model once the latter is written on the logit scale — is it still needed?*

Good catch, it is not: once the assignment model is written directly on the logit scale, the calibration equation collapses to the same closed form, so we have removed it as redundant.

Removed: the closed-form calibration equation, which became redundant once sign-off choice was modeled directly on the logit scale above.

**Reviewer 1 — comment 7**
*An analysis of robustness to unmeasured confounding is missing — sender urgency in particular seems like an obvious candidate.*

We agree this was a gap and have added a sensitivity analysis for unmeasured caffeine confounding, cited in the manuscript, as a proxy for sender urgency. (modified on p. 3)

**Reviewer 1 — comment 8**
*The conclusion's framing of the emoji sign-off as "not a costless substitute" is memorable but should probably be tied more explicitly to the effect estimates — or else softened.*

We have reviewed this passage and, on balance, prefer to keep the wording as originally written.

**p. 3** — Taken together, these findings suggest that a lone "👍" emoji, while efficient, may not be a costless substitute for a written sign-off in professional correspondence.

**Reviewer 1 — comment 9**
*Please also report sensitivity to the email client used, in case client-side rendering of the emoji affects the outcome.*

This analysis was performed but, on reflection, is redundant with the caffeine-confounding sensitivity analysis and has been removed for concision.

Removed: the sensitivity analysis by email client.

**Reviewer 1 — comment 10**
*One comment, filed here for convenience even though it concerns two separate locations in the manuscript: the operational definition of "sign-off," introduced early on, should be explicitly reaffirmed in the conclusion.*

Addressed in both locations: (modified on p. 1 and p. 3).

Shown as two separate excerpts, one per location:

**p. 1** — This comment required a correction in two places in the manuscript — this sentence, in the introduction, and a matching one in the conclusion — applies equally to both instances.

**p. 3** — As already noted in the introduction, this conclusion applies equally to the paired instance of the same reviewer comment discussed there.

## Reviewer 2

**Reviewer 2 — comment 1**
*The "Statistical analysis" heading should indicate that the analysis plan changed relative to what was originally registered.*

The heading has been updated accordingly.

**p. 2** — Statistical analysis (updated)

**Reviewer 2 — comment 2**
*A subgroup analysis (recipient seniority, day sent, CC'd manager, email length) would help readers judge how broadly the main finding generalizes.*

A subgroup figure has been added, reporting the requested comparisons. (modified on p. 2)

**Reviewer 2 — comment 3**
*The "🙏" null-comparator sub-analysis is an unusual choice and, frankly, distracts from the paper's main contribution — consider removing it given the space constraints.*

We agree and have removed this sub-analysis in its entirety, including its figure and table.

Removed: the "🙏" null-comparator sub-analysis, including its figure and table.

**Reviewer 2 — comment 4**
*The discussion's causal language ("causes an increase") is stronger than the observational design supports, IPW notwithstanding.*

We have softened this to an associational claim and flagged residual confounding by unmeasured sender urgency, in the same spirit as our response to reviewer 1, comment 2, p. 1 on accounting for measured confounders.

**p. 3** — ~~These results suggest that replying "merci" causes an increase in the probability of a further reply, relative to a "👍" emoji.~~ These results are consistent with an association between sign-off choice and the probability of a further reply, though residual confounding by unmeasured aspects of sender urgency cannot be ruled out.

**Reviewer 2 — comment 5**
*Using battery level as an instrument seems like a stretch — please justify the exclusion restriction more carefully, or drop this analysis.*

We have added an explicit statement of the exclusion restriction and a citation to prior work on battery-level instruments in applied econometrics; we believe the assumption, while unusual, is no less defensible than in other applications of the design.

(modified on p. 2)

**Reviewer 2 — comment 6**
*The reply-type table should include a "reacted only, no reply" category — a thumbs-up reaction with no further text is common enough in our experience to warrant its own row.*

Added, at 7% of threads by the 48-hour mark.

**p. 3** —

| Reply type | By 24h | By 48h |
| --- | --- | --- |
| Full-text reply | 29% | 38% |
| One-word reply | 11% | 14% |
| Emoji-only reply | 6% | 9% |
| Reacted only (no reply) | – | 7% |

Table II: Reply type breakdown, by 24 and 48 hours after the final message.

### Editor

**Editor — comment 1**

*Please ensure that every figure and table added or modified in response to the reviewers is cited with a page number in this letter, per journal policy.*

Done throughout; see in particular the subgroup figure, referenced at Fig. 2, p. 2, and the reply-type table, referenced at Table II, p. 3.

## Author notes

The changes above address every reviewer and editor comment. Two further, smaller changes were made on our own initiative during revision, noted here for transparency even though neither answers a specific reviewer comment.

### Marcus Cc

**Marcus Cc — change 1**

While implementing the change requested in our response to reviewer 1, comment 2, p. 1, I checked covariate balance after weighting and added a note confirming adequate overlap.

**p. 2** — While implementing the change above, we also checked covariate balance across sign-off groups after weighting and found adequate overlap, with no standardized mean difference exceeding 0.1.

### Constance Reply-All

**Constance Reply-All — change 1**

Re-read the sign-off definition after the change above and confirm it still matches our extraction procedure.

(see p. 1)

## For the reviewers only

The table below summarizes, for the reviewers' convenience only, how many comments from each reviewer led to an actual textual or figure/ table change in the manuscript, as opposed to a clarification with no change — consistent with prior work on the sociology of email reply latency T. Sloe [3].

| Reviewer | Comments | Led to a change |
|---|---|---|
| Reviewer 1 | 10 | 8 |
| Reviewer 2 | 6 | 6 |

Table R3: Summary of comments and resulting changes, for the reviewers' convenience.

As shown in Table R3, the large majority of comments from both reviewers led directly to a manuscript change, consistent with our own prior finding that inbox behavior is, if nothing else, a reliable trigger for further action C. Reply-All [4].

## References cited in this response

[1] M. Cc and P. Bcc, "Norms of the Professional Email Sign-Off: A Corpus Study," *Annals of Workplace Linguistics*, 2021.

[2] I. Weightley, "Inverse Probability Weighting for Non-Randomized Communication Trials," *Journal of Applied Statistics*, 2019.

[3] T. Sloe, "The Sociology of Email Reply Latency," *Journal of Irreproducible Snacking*, 2018.

[4] C. Reply-All, “Reply-All Fatigue: A Field Study of Inbox Behavior,” *Journal of Organizational Communication*, 2020.